

FRANKLINE MISANGO OYOLO

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SUMMARY

An Aspiring Doctoral candidate with research interests in Artificial Intelligence. Current research experience in spatio-temporal deep learning models and diffusion convolutional architectures alongside applied scientist exposure to Data centric industrial roles

EDUCATION

The University of Hong Kong

Expected Jun 2027

BEng. Artificial Intelligence & Data Engineering — GPA: 3.73/4.3

Hong Kong SAR

- **Relevant courses:** Robotics, Partial Differential Equations, Machine Learning, Probability & Statistics, NLP, Scientific Computing, Analysis of Algorithms, Operating Systems, MATLAB, Computer Architecture, Discrete Mathematics

RESEARCH EXPERIENCE

Swiss Federal Institute of Technology Lausanne (EPFL)

Jul 2025 – Aug 2025

Summer@EPFL PhD Research Assistant - Lab for Media and Design

Lausanne, Switzerland

- Co-authored a peer-reviewed study introducing a Diffusion Convolutional Recurrent Neural Network (DCRNN) for forecasting bike-sharing OD flows across Switzerland.
- Engineered a multi-scale feature integration framework using OpenStreetMap and population data at 500m–1500m radii to encode hierarchical urban context for 1,000+ stations.
- Formulated the task as sequence-to-sequence learning with DCGRU cells to capture spatial diffusion and temporal dependencies.
- Built a 10M+ record dataset combining trip logs, OSM features, and population grids for robust model training.
- Achieved 27% RMSE reduction vs. ST-GCN and 13.4% gain from OSM integration, validated via ablation studies.
- Performed attention-based interpretability analysis showing urban stations attend to local features, while peripheral ones emphasize regional connectivity.
- Developed a geospatial ingestion pipeline integrating 3 heterogeneous data sources with 99.7% accuracy.
- Deployed containerized inference endpoints on EPFL's HPC cluster, reducing latency by 40%.

Department of Architecture, University of Hong Kong

Dec 2023 – Apr 2024

PhD Research Assistant (AI)

Hong Kong SAR

- Engineered Meta SAM transformer-based instance segmentation pipeline (ViT-H backbone, 632M parameters) achieving 91.8% IoU on 1000+ architectural floor plans for aeration feature analysis in microbial containment studies
- Implemented distributed coordination algorithm for 200-node swarm robotics system using ROS2 Galactic and ZeroMQ pub/sub architecture, with <5ms latency for real-time airflow monitoring across 100+ housing units
- Fine-tuned DeepLab 3+ with Xception-65 backbone (26.5M parameters) on segmented thermal imagery dataset (n=4,800) for temporal prediction of ventilation efficiency, achieving 0.87 R² on 3-year forecasts
- Architected Kubernetes-orchestrated MLOps pipeline with COCO annotation standard, reducing data preprocessing time by 78% and enabling automated multi-GPU training across 3 distributed DL workloads

Center for Teaching and Learning, University of Hong Kong

Mar 2023 – May 2023

PhD Research Assistant (Advanced Analytics)

Hong Kong SAR

- Implemented BERT-Large (340M parameters) with custom tokenization pipeline for semantic analysis of 30k+ student responses, extracting 128-dimensional embeddings with 94.2% topic classification accuracy
- Developed Meta BART-based bidirectional transformer sentiment analysis system with attention visualization, achieving F1 score of 0.89 on 5-point sentiment scale across 17 pedagogical categories
- Engineered distributed data lake architecture (S3-compatible object storage with 12TB capacity) processing 1000+ survey datasets with automated ETL pipelines, enabling sub-second query response for longitudinal analytics

WORK EXPERIENCE

BASF SE

Jan 2025 – Jun 2025

Artificial Intelligence Engineer - Digital Business Solutions

Hong Kong

- Architected real-time supply chain dashboard using React/Redux with TensorFlow.js LSTM models (91% forecast accuracy) processing 14TB+ logistics data, reducing stockouts by 38% through automated replenishment signals
- Optimized Spark ETL pipelines using Parquet columnar storage & Airflow dynamic task mapping, achieving 88% memory efficiency (2.4× throughput) on 64-node EMR cluster handling 500GB/day shipments
- Engineered hybrid search system combining Azure Cognitive Search with custom BERT-As-Service (96.2% NDCG@10), reducing NLP query latency 42% via FPGA-accelerated ONNX runtime deployments

Manulife Financial Corporation

Jun 2024 – Aug 2024

Data Governance Summer Analyst

Hong Kong

- Developed enterprise data catalog using React/Node.js with ArangoDB graph backend, achieving 87.2% automated classification of 2.3M+ assets through TF-IDF/Word2Vec similarity scoring at 1.2k QPS
- Redesigned Spring Boot API layer using concurrent Neo4j traversals & Hazelcast caching (40% latency reduction), handling 450 RPS with P99 of less than 120ms on 16-core Azure instances
- Implemented testing framework with Jest/Cypress & Jenkins pipeline integration (98% coverage), catching 142 critical bugs pre-production through mutation testing & chaos engineering scenarios

TEACHING ASSISTANTSHIPS

The University of Hong Kong
ENGG1340/COMP2117 Student TA

Sep 2024 - Dec 2024
Hong Kong SAR

- Architected multi-AZ AWS VPC infrastructure (t3.medium EC2 instances, S3 storage, RDS PostgreSQL) to support 10-week programming assessments with CI/CD pipeline for Bash scripting and C++14 standard compiler integration
- Implemented rubric-based assessment framework for evaluating 100+ weekly submissions against 17-point code quality metrics, including algorithmic complexity ($O(n)$ analysis), memory management, and SOLID principles
- Mentored 100 students through terminal-based game development projects utilizing POSIX-compliant shell scripting and C++ STL implementations with full project completion rate

PUBLICATIONS

- [1] F. M. Oyolo (HKU), J. Coquet (ETH Zurich), and J. Huang (EPFL), "Attention-Fused Multi-Scale OSM Context for Spatio-Temporal GNNs in Urban Mobility Flow Forecasting," ACM SIGSPATIAL 2025, Nov 2025.
- [2] F. M. Oyolo (HKU) and Robert J. Otieno (Caltech), "Cross-Regime Analysis of Leveraged Portfolio Strategies," Caltech Quantitative Finance Society Association. 
- [3] F. M. Oyolo, "Market Making Through Jump Diffusion HJB Equations with Order Flow Toxicity Tracking," HKU Trading Group. 

HONORS & AWARDS

Mar 2025: Granted the PyconUS Travel Grant by the Python Software Foundation to attend the PyCON in Pennsylvania
Jan 2025: Awarded the Summer @ EPFL Grant of CHF 5000 of a 1.6% competitive rate globally to complete the SURF
Dec 2024: Granted The All Builders Welcome Grant by Amazon to attend the AWS Re:Invent conference in Las Vegas
Apr 2023: Awarded \$5000 for winning the AI Tech4good Hackathon organized by Accenture
Sep 2022: Awarded \$10,000 as African Impact Grant Award for winning the Healthcare theme by Mastercard Foundation
Aug 2022: Granted \$150,000 for 4 years, termed Future Leaders Scholarship by HKU, to study Engineering
May 2022: Granted CHF 4000 by Glencore to attend the Swiss-African Hydrogen case challenge in Switzerland

LANGUAGES, INTERESTS & ACTIVITIES

Languages: English (Native), Swahili(Native), Luhya (Native), German (B1 Fluency), French (A1 Fluency)
Interests: Chess (Blitz 1800), Leetcode (top 5%), Cycling, International travel, Reading on Financial Markets and Global Politics
Campus Leadership: HKU Trading Group TMT Software Developer, HKU Quantitative Researcher for Analysis of Algorithms
Certifications: AWS Machine Learning Specialty, AWS Solutions Architect Professional

TECHNICAL PROJECTS

AlgoForge | *Python, Transformers, LLM Engineering* 2025

- Architected multi-model NLP pipeline comparing BERT (84.2% F1), BART (89.1% F1), and Llama-2-70B for financial document analysis, processing 50K+ news articles/day from 37 sources using HuggingFace Transformers and FAISS vector search
- Implemented LoRA fine-tuning with 8-bit quantization (QLoRA), reducing BART model size by 63% while maintaining 98.7% of original accuracy on FiQA benchmark dataset
- Developed custom attention masking for financial jargon handling, improving rare term (frequency $\geq 0.1\%$) sentiment accuracy by 41% compared to baseline BERT architecture

SERSi Spacecraft Algorithm | *Python, NASA Data, Planetary Science* 2025

- Devised signal reconstruction pipeline leveraging wavelet-based denoising (Daubechies-4) with Maxwell-Boltzmann distribution fitting to recover radiation-corrupted spectroscopic albedo measurements, reducing data loss by 78.3% across 87TB of iPAC/TESS multi-spectral imagery
- Formulated Bayesian uncertainty quantification framework with MCMC sampling ($n=10^7$ iterations) for exoplanet habitability indices, achieving AUC=0.94 on ESI prediction across 1,400+ KOI candidates with 27 spectral features

DeepRacer | *Python, Reinforcement Learning, ROS2, AWS* 2024

- Developed modular reinforcement learning architecture with dynamic action space definitions and reward function templates, achieving 8.4s/lap times through hyperparameter-optimized training (256-episode batches) using AWS SageMaker and ONNX runtime deployment
- Implemented track-specific path planning with Bezier curve optimization, enabling 72km/h straightaways and 45° turn negotiation through progressive steering PID control ($K_p=0.7$, $K_i=0.01$, $K_d=0.03$), ranked top 5% in AWS Global Virtual Circuit among 14K teams
- Designed distributed training system using deepracer-on-the-spot framework with dual-node architecture (Nginx load balancing), processing 18TB training data through automated log analysis pipelines with 92% sample efficiency

BRH35 Brain Cancer Detection | *Flask, CNN, Transfer Learning* 2024

- Constructed distributed Flask microservice architecture (Gunicorn WSGI, 12-worker Nginx load balancing) with MobileNetV3-Large backbone (5.4M parameters) fine-tuned on 3D MRI volumetric data achieving 94.7% classification accuracy with 18ms inference latency
- Integrated HL7-compliant PACS integration layer using DICOM networking protocol (C-STORE SCP) with lossless JPEG2000 compression, enabling 14TB/day throughput with automated lesion segmentation via U-Net architecture (IoU 0.89)